Automated Water Quality and Contamination Detection

GROUP RESEARCH THESIS — INTEGRATED DESIGN DRAFT

Team member	Student ID	Individual component
W.A.S.P.D. WIJESINGHA	IT22295460	Visible contamination detection
W.N.N. DE SILVA	IT22211750	Water quality classification
D. DODANWELA	IT22212122	Microorganism detection
R.M.N.A. RATHNAYAKE	IT22193940	Water safety report and purification advisor

Research design draft — experiments, team identities and sign-off pending

Bachelor of Science Honours in Information Technology

Specialized in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

October 2026

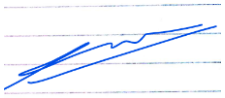
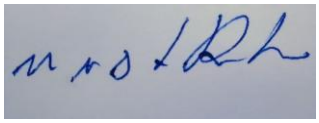
DECLARATION AND SUPERVISOR CERTIFICATION

Candidate declaration pending signature. The candidate declaration and any reproduction permissions must be reviewed and confirmed by all four students before submission. Names, student IDs and signatures have not been supplied; no declaration is represented as signed or approved in this draft.

Academic integrity and AI disclosure. Generative AI assisted in drafting and organising this report from the supplied project descriptions. The students must review the technical claims, references and proposed methods, document any further AI use, and confirm compliance with the applicable module policy before submission.

Student name	Student ID	Signature	Date
W.A.S.P.D. WIJESINGHA	IT22295460		11/10/2026
W.N.N. DE SILVA	IT22211750		11/10/2026
D. DODANWELA	IT22212122		11/10/2026
R.M.N.A. RATHNAYAKE	IT22193940		11/10/2026

Supervisor certification pending. Supervisor and co-supervisor names, certification, signatures and dates must be completed by the authorised individuals.

Role and name	Signature	Date
Supervisor name: Mr. Buddika Harshanath		11/10/2026
Co-supervisor name: Prof. Samantha Rajapaksha		11/10/2026

ABSTRACT

This report proposes an integrated system for assessing water samples using visible images, physicochemical measurements and microscope imagery. The common problem is the difficulty of combining these different forms of evidence into a traceable assessment that users can understand. Four components address this problem: computer vision for visible contamination detection; machine learning for water quality classification; deep learning for microorganism detection; and a generative AI reporting and purification advisor supported by retrieval-augmented generation. A shared sample identifier connects the inputs and results. An aggregation layer preserves parameter warnings, uncertain detections and missing data before a report is generated from approved reference material. The proposed methodology includes expert annotation, grouped training and test splits, comparative model evaluation, interface testing and stakeholder review. Evaluation will measure detection performance, unsafe-class recall, report grounding, handling of incomplete samples and end-to-end response time. The supplied documents establish the functional design but contain no measured experimental results or deployment evidence. Consequently, this report presents the integrated research design and validation plan rather than a completed experimental study. Its intended contribution is a coordinated decision-support workflow that preserves the origin and limitations of each finding. Demonstrating reliable performance on representative samples and obtaining expert review are necessary before operational use.

Keywords: water quality, object detection, machine learning, microscopy, retrieval-augmented generation

ACKNOWLEDGEMENTS

We would like to express my sincere gratitude to my supervisor, Mr. Buddika Harshanath, and my co-supervisor, Prof. Samantha Rajapaksha, for their guidance, constructive feedback and encouragement throughout this research. We also thank the Department of Information Technology at the Sri Lanka Institute of Information Technology (SLIIT) for the academic support and opportunity to undertake this study. We appreciation extends to our research group members for their collaboration, shared ideas and support in developing the overall water-quality assessment project. Their contributions have helped shape my individual component, the Water Safety Report and Purification Advisor. Finally, We are grateful to my family and friends for their patience, understanding and continued encouragement throughout our academic journey.

TABLE OF CONTENTS

DECLARATION AND SUPERVISOR CERTIFICATION i
ABSTRACT ii
ACKNOWLEDGEMENTS iii
LIST OF FIGURES vii
LIST OF TABLES viii
LIST OF ABBREVIATIONS ix
1 INTRODUCTION AND PROJECT CONTEXT 1
1.1 Background and context 1
1.2 Stakeholders and user needs 1
1.3 Alignment with the Sustainable Development Goals 1
1.4 Purpose and scope of the common report 2
1.5 Background Literature and Theoretical Foundations 2
1.6 Critical Review of Relevant Approaches 3
1.7 Research Position and Claimed Novelty 4
2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS 6
2.1 Overall problem definition 6
2.2 Root causes and overall research gap 6
2.3 Subproblem and component mapping 6
2.4 Overall aim and group objectives 7
2.5 Scope assumptions and constraints 7
2.6 Testable Research Questions and Hypotheses 8
2.7 Operational Boundaries and Risk Controls 9
3 OVERALL SOLUTION STRUCTURE 10
3.1 Integrated solution overview 10
3.2 High level architecture 10
3.3 Component 1 summary 11
3.4 Component 2 summary 12
3.5 Component 3 summary 12
3.6 Component 4 summary 12
3.7 End to end user and data flow 13
3.8 Data Acquisition, Sample Identity and Provenance 13
3.9 Label Taxonomy and Annotation Governance 14
3.10 Dataset Partitioning and Leakage Prevention 15
3.11 Model Selection and Baseline Comparisons 15
3.12 Integrated Decision Semantics 16
3.13 Retrieval, Reporting and Chemical Calculation Controls 18
4 COMPONENT INTEGRATION AND COHESIVE OPERATION 20

4.1 Integration strategy 20
4.2 Interfaces data contracts and shared models 20
4.3 Shared nonfunctional requirements 20
4.4 Integrated workflow and exception handling 21
4.5 Team contribution and dependency matrix 21
4.6 API Contract and Cross-component Validation 22
4.7 Orchestration, Retry and Failure State Design 23
4.8 Auditability and Reproducibility Requirements 24
5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION 25
5.1 Common technology stack and infrastructure 25
5.2 Development and integration process 25
5.3 Deployment and operational configuration 25
5.4 Evidence of team execution 26
5.5 Development Framework and Verification Strategy 26
5.6 Deployment Topology and Operational Security 27
5.7 Professional Practice, Accessibility and Ethics 28
6 INTEGRATED TESTING EVALUATION AND VALIDATION 29
6.1 Evaluation framework and success criteria 29
6.2 Integration and end to end testing 29
6.3 Overall results 30
6.4 Stakeholder and user validation 30
6.5 Achievement of common objectives 31
6.6 Integrated limitations and threats 31
6.7 Component-level Evaluation Protocols 31
6.8 Integrated Validation and Baseline Comparison 32
6.9 Statistical Reporting and Robustness Analysis 33
6.10 Results Reporting and Evidence Register 34
7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION 36
7.1 Value proposition and beneficiaries 36
7.2 Market alternatives and differentiation 36
7.3 Business model and feasibility 36
7.4 Adoption intellectual property and compliance 37
7.5 Sustainable Development Goal contribution 37
7.6 Commercialisation Validation and Positioning 37
7.7 Preliminary Sustainability and Cost Framework 38
8 CONCLUSION AND FUTURE DIRECTIONS 40
8.1 Overall conclusion 40

8.2 Principal integrated contribution 40
8.3 Future directions 40
8.4 Synthesis of Group-level Contributions 40
8.5 Prioritised Completion Plan 41
REFERENCES 42
APPENDICES 44
Appendix A Integrated interface specification 44
Appendix B Team contribution declaration 45
Appendix C Integrated test matrix 45
Appendix D: Shared Data Dictionary 46
Appendix E: Expanded Fault-Injection and Security Test Plan 46
Appendix F: Dataset and Model Reporting Checklist 47
Appendix G: Academic Integrity and AI-use Disclosure 47

LIST OF FIGURES

Figure 3.1 Proposed integrated system architecture 11
Figure 3.2: Integrated sample lifecycle and report release (proposed design; not experimental evidence). 17
Figure 3.3: Logical data entities and evidence relationships (proposed design; not experimental evidence). 18
Figure 3.4: Retrieval-supported reporting with claim-checking safeguards (proposed design; not experimental evidence). 19
Figure 4.1: Analysis state machine and fail-safe routing (proposed design; not experimental evidence). 24
Figure 6.1: Proposed grouped evaluation and validation protocol (proposed design; not experimental evidence). 33

LIST OF TABLES

Table 1.1: Literature and technique comparison for the proposed multi-modal system 4
Table 2.1 Subproblems and component responsibilities 7
Table 4.1 Proposed contribution and dependency matrix 21
Table 4.2: Proposed shared API contract fields 22
Table 6.1 Required results and current evidence status 30
Table 6.2: Required empirical evidence before final submission 34
Table C.1 Planned integrated test cases 45

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial intelligence
API	Application programming interface
CNN	Convolutional neural network
IoT	Internet of Things
JSON	JavaScript Object Notation
LLM	Large language model
mAP	Mean average precision
ML	Machine learning
RAG	Retrieval-augmented generation
REST	Representational state transfer
SDG	Sustainable Development Goal
TDS	Total dissolved solids
WHO	World Health Organization
YOLO	You Only Look Once

1 INTRODUCTION AND PROJECT CONTEXT

1.1 Background and context

Water quality assessment involves several kinds of evidence that answer different questions. A photograph records visible appearance, laboratory or sensor measurements describe selected physicochemical properties, and microscopy can support the investigation of organisms. The proposed project combines these inputs in one sample record so that a reviewer can inspect both the assessment and its supporting evidence [1], [2].

The World Health Organization describes drinking-water quality management as a risk-based process extending from catchment to consumer, supported by health-based targets and independent surveillance [3]. This project is positioned as research into decision support within that wider process. A favourable prediction from a limited set of inputs will not be treated as certification that a sample meets every drinking-water requirement.

1.2 Stakeholders and user needs

The intended primary users are laboratory technicians, water quality officers and trained environmental monitoring staff. They need clear input requirements, traceable sample records, interpretable warnings and a way to refer uncertain findings for review. Community water operators and household users are potential recipients of simplified reports, while researchers and supervisors require reproducible methods and auditable evaluation evidence.

These needs are design assumptions derived from the proposed workflow. Stakeholder interviews and observations have not been supplied. The team will confirm the intended use setting, available instruments, acceptable reporting language and escalation procedures before finalising requirements. Laboratory users should be involved in defining sample handling and reference labels.

1.3 Alignment with the Sustainable Development Goals

The project is intended to support Sustainable Development Goal 6, particularly target 6.1 on access to safe and affordable drinking water and target 6.3 on improving water quality [4]. Its potential contribution is better organisation and

interpretation of monitoring evidence. Any contribution would depend on reliable measurements, appropriate follow-up and access to suitable treatment. No improvement in population health, water access or pollution reduction has yet been demonstrated.

1.4 Purpose and scope of the common report

This common report describes the shared problem, relationships among the four components, system architecture, interfaces, evaluation plan and proposed route to use. Detailed datasets, experiments and model development belong in the individual component theses. The two supplied project documents provide the functional scope [1], [2]; the API contracts, acceptance criteria and development arrangements below are proposed elaborations.

The available evidence is limited to design descriptions. Student identities, degree details, datasets, implementation records, stakeholder findings and measured results remain unverified. Sections on execution and evaluation therefore distinguish planned activities from demonstrated achievements.

1.5 Background Literature and Theoretical Foundations

The evaluation of drinking-water quality is fundamentally different from deciding whether a single image looks clear. Risk management requires a defined use context, appropriate sampling, adequate identification of hazards, and surveillance across the system from source to consumer. WHO drinking-water guidance describes preventive water-safety planning and independent surveillance, making it an appropriate reference for the interpretation and limits of this prototype [3]. The proposed research therefore separates individual sensor or image observations from claims about full drinking-water safety.

Computer vision and machine learning provide different forms of evidence. Object detection models such as YOLO predict locations and classes in images, while region-proposal approaches such as Faster R-CNN provide an alternative accuracy–complexity trade-off [5], [6]. Tabular classifiers infer class labels from numerical variables, whereas a deterministic rule engine checks explicitly documented parameter criteria. Combining these approaches is not automatically a validation: the

utility of multi-modal assessment must be demonstrated on linked samples, with each stream independently checked and its uncertainty retained.

Retrieval-augmented generation (RAG) joins language generation to search over an external collection of evidence [8]. This is valuable for making structured findings readable and traceable, but passage retrieval alone does not guarantee factual accuracy. The report layer must verify source relevance, distinguish quoted guidance from model output, and avoid converting incomplete observations into an unqualified safe/unsafe claim. Household treatment performance, especially microbial removal, requires testing against defined health-based criteria rather than relying on a text recommendation [9].

1.6 Critical Review of Relevant Approaches

A visual-only workflow is relatively easy for users to operate but observes only surface or camera-visible properties. Its findings depend heavily on lighting, container colour, camera distance, turbidity, reflection, and training representation. Objects such as floating debris may have clear spatial boundaries; oil sheen, diffuse discolouration, and appearance associated with turbidity may not. A rigorous study must decide which labels are object-detection targets, which are image-level appearance classes, and which are better measured instrumentally before choosing a network architecture.

Numerical water classification provides a complementary view but introduces its own limitations. Water with similar pH or dissolved-solids values may differ in chemical composition, biological hazards, or intended use. A model trained on a convenience dataset can learn correlations that are absent in deployment. Therefore the research design requires provenance for every training label, separation of samples between data partitions, parameter units, comparison against simple documented rules, and reporting of the unsafe-class false-negative rate rather than overall accuracy alone.

Microscopic imagery is a third distinct data domain. Organism identity depends on acquisition method, staining or sample preparation, resolution, expert reference assessment, and the limits of visual morphology. A system trained to detect broad organisms must not claim species-level confirmation when the reference standard

does not support that specificity. For instance, object counts in a field of view cannot be reported as concentration per millilitre without calibrated sample volume and field geometry. These limitations shape both annotation policy and final report wording.

A fourth group of related approaches concerns explanation and automated advice. Generic language models can write coherent summaries without evidence for a specific claim. RAG improves access to controlled knowledge, but introduces retrieval relevance, outdated guidelines, citation mismatch, and prompt-injection risks. The project proposes deterministic dose calculations only as a separate, validated function with explicit input requirements; the generation model is never authorised to invent treatment amounts. The architecture aligns with transparent dataset documentation and model reporting recommendations [10], [11].

Table 1.1: Literature and technique comparison for the proposed multi-modal system

Technique	Useful contribution	Principal limitation	Planned response
Image object detection [5], [6]	Localises visible objects	Acquisition bias; diffuse phenomena	Controlled annotation and holdout
Tabular classification [7]	Learns combinations of readings	Label drift and hidden hazards	Rules baseline and unsafe recall
Microscopy detector [5], [6]	Locates annotated organisms	Expert identification limits	Taxonomy constrained by evidence
RAG [8]	Grounded language generation	Incorrect retrieval or unsupported claims	Passage-level citation checking
WHO treatment guidance [3], [9]	Risk and performance context	Not itself a sample measurement	Human review and treatment safeguards

1.7 Research Position and Claimed Novelty

The proposed contribution is not a claim that each constituent algorithm is new. Rather, the research tests whether one versioned sample record can preserve the identity, acquisition conditions, confidence, missingness, and contradictory outputs of three specialised analyses before generating a transparent report. The integration questions are distinct from achieving a high score on three unrelated datasets. A completed research evaluation must establish whether this fusion reduces unsupported claims or improves interpretation compared with simply presenting four independent screens.

The research gap remains provisional until a structured review establishes the degree to which closely related systems already address the same problem. Prior art should be compared on modalities, linked-sample validation, annotation reference standards,

evidence provenance, incomplete-data handling, human oversight, and treatment-advice controls. Novelty will be justified through the measurable combination of these characteristics in the specified environment, not through unverified universal statements such as being the first water-quality AI system.

2 OVERALL PROBLEM DEFINITION AND FOUR SUBPROBLEMS

2.1 Overall problem definition

The common problem is that users need a coherent assessment of a water sample while the relevant evidence is distributed across visible observations, physicochemical measurements and microscopic findings. Without a shared record and explicit interpretation rules, evidence may be incomplete, inconsistently labelled or difficult to explain. The research question is whether a coordinated workflow can integrate these modalities into a more traceable and understandable assessment while preserving uncertainty and avoiding unsupported safety claims.

2.2 Root causes and overall research gap

The design addresses three integration difficulties: each input has a different format and quality requirement; findings are produced at different times and levels of certainty; and recommendations require context beyond a model label. For example, a negative image detection is a result about an image, while a parameter measurement is a value with a unit, method and sampling time. Treating these outputs as interchangeable would lose information.

The proposed research gap is an integration and validation question within the selected use setting. The supplied documents do not establish that no similar system exists. A systematic literature and product review is still required before claiming novelty. This report focuses on whether the proposed combination adds measurable value over separate component reports and a rules-only assessment.

2.3 Subproblem and component mapping

The four subproblems are deliberately separated by input and responsibility. Component 1 interprets visible images; Component 2 interprets measured numerical parameters; Component 3 interprets microscope images; and Component 4 explains the combined evidence using retrieved guidance. Table 2.1 maps these responsibilities. Shared aggregation supports the four components and is not an additional research component.

Table 2.1 Subproblems and component responsibilities

Component	Subproblem	Evidence passed forward
1 Visible detection	Locate visible contamination	Image regions and quality flags
2 Quality classification	Interpret measured parameters	Category and parameter warnings
3 Microorganism detection	Locate validated organism classes	Image detections and counts
4 Report and advisor	Explain combined evidence	Cited report and review needs

2.4 Overall aim and group objectives

The overall aim is to design and evaluate a traceable, four-component water quality decision-support system. The group objectives are:

O1: Detect and localise the agreed visible contamination classes, measuring per-class precision, recall and mean average precision on a held-out image set.

O2: Classify measured water quality using a documented target definition, comparing Random Forest and Gradient Boosting with a rules-only baseline and measuring unsafe-class recall and macro F1.

O3: Detect agreed microorganism categories in expert-labelled microscope images, measuring per-class recall, localisation quality and count error where count labels are reliable.

O4: Produce reports linked to findings and approved references, evaluating citation support, factual consistency, missing-data handling and expert acceptability.

O5: Integrate all components under a shared sample identifier and demonstrate correct association, exception handling and reproducibility through the test cases in Appendix C. All objectives remain subject to experimental verification.

2.5 Scope assumptions and constraints

The research scope includes image upload, numerical data entry or import, model inference, evidence aggregation and report generation. Initial operation is assumed to involve trained users and access to laboratory measurements and microscopy. Dataset access, permissions, annotation effort, hardware capacity and expert availability are dependencies.

The scope excludes autonomous treatment control and a claim that photographs alone establish potability. Microplastic identification requires a validated acquisition and reference method and is outside the initial ordinary-photograph scope. Bacterial species labels will only be used if the imaging protocol and independent reference labels support that distinction; otherwise the model will report broader validated categories. Unmeasured chemical or biological hazards remain outside the assessment.

2.6 Testable Research Questions and Hypotheses

RQ1 asks whether object-detection performance for the agreed visible contamination categories generalises to held-out sampling conditions. RQ2 asks whether a calibrated multi-parameter classifier detects unsafe-labelled records more reliably than explicit parameter rules. RQ3 asks whether the microorganism detection model can reproduce expert annotations at the stated taxonomic level. RQ4 asks whether retrieval-supported reporting reduces unsupported treatment claims and improves traceability compared with a non-retrieval template. RQ5 asks whether combining modalities under a shared sample identifier reduces association errors and ambiguous assessments.

Each question demands a matching comparator and independently assessed target. For RQ1 and RQ3, the test unit must remain the sample or acquisition session rather than a random image crop. For RQ2, the source of safe/unsafe labels must be documented; an arbitrary threshold-derived label would unfairly favour the rules baseline. For RQ4, human raters should evaluate blinded reports for supported claims, missing evidence, and clarity. For RQ5, injected integration faults should produce explicit error states instead of apparently normal output.

The project should preregister or otherwise record its principal metrics and acceptance thresholds before examining the final test set. Where formal hypothesis testing is appropriate, confidence intervals and paired comparisons should accompany observed differences. A result that fails a hypothesis can still be a useful research contribution when it reveals a defensible system boundary or a reproducible error mode.

2.7 Operational Boundaries and Risk Controls

The project distinguishes water-type classification, selected parameter screening, microorganism image detection, and certification of drinking-water safety. These are not equivalent outcomes. Water can appear clear and fall within measured parameter ranges while still containing an unmeasured hazardous chemical or pathogen. Reports should say exactly which modalities were assessed and whether mandatory measurements are absent. A favourable component classification is therefore described as “no concern detected within assessed scope” rather than an unconditional declaration of potability.

The proposed chemical-dose calculator is a research module with a restricted input schema, reviewed formulas, explicit units and clearly defined supported conditions. It must not provide autonomous instructions for use when source characteristics, available products, required residual monitoring or expert approval are missing. Any future treatment performance claim would require independent laboratory validation against applicable guidelines [3], [9]. This separation is essential both for research validity and for responsible communication to non-specialist users.

3 OVERALL SOLUTION STRUCTURE

3.1 Integrated solution overview

The system will maintain one evidence record for each water sample. Three analytical services will independently process the modalities available for that record. A shared aggregation service will check completeness, retain warnings and attach model and data versions. Component 4 will then produce a report from the structured findings and retrieved reference passages.

The central value proposition is traceability: a reviewer should be able to move from a report statement to the measurement, image region or reference that supports it. The design will also distinguish completed, failed and unavailable analyses so that missing evidence cannot silently appear as a favourable result.

3.2 High level architecture

Figure 3.1 shows the proposed architecture. The authenticated interface and orchestration service control sample identity and access. Images are stored separately from structured results. The three analytical branches may execute independently, but report generation uses a defined snapshot of their results. The approved knowledge base is curated from external references; retrieved text is treated as evidence rather than executable instructions.

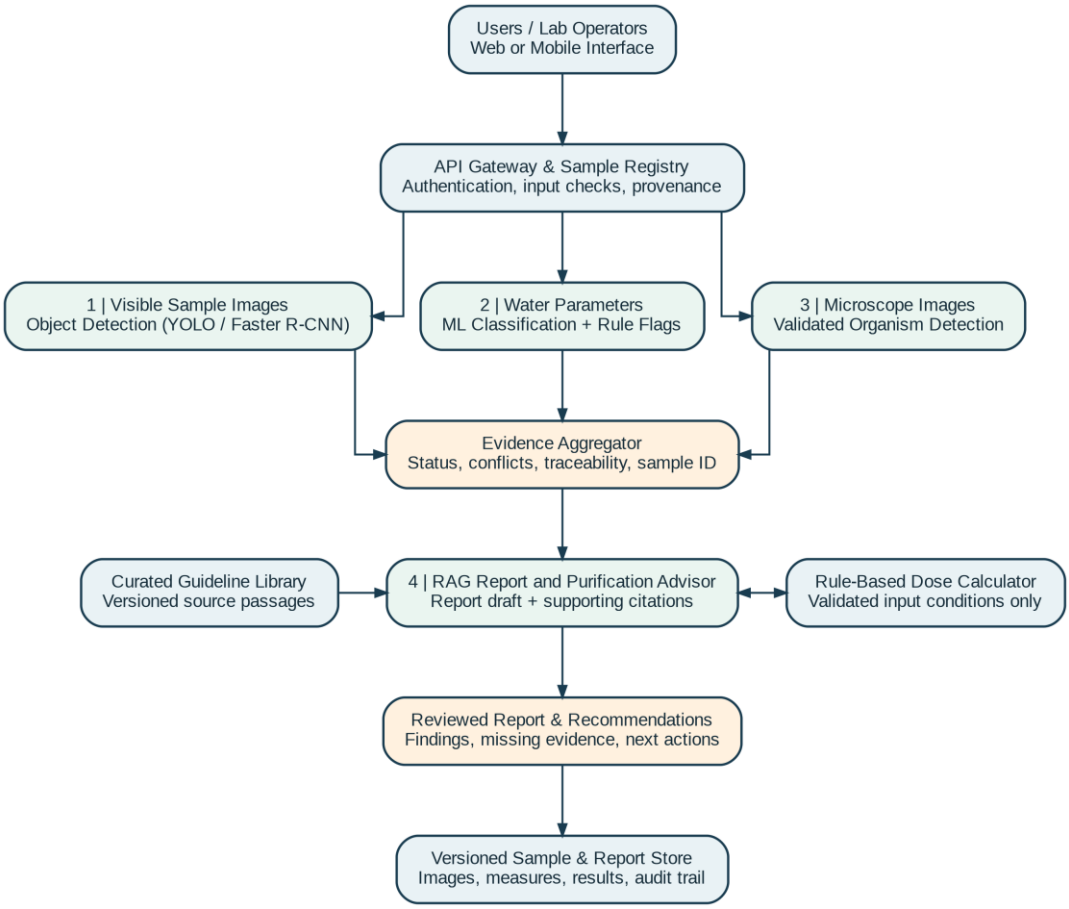


Figure 3.1 Proposed integrated system architecture

3.3 Component 1 summary

Proposed owner: Component 1 student, identity pending. This component addresses visible contamination using camera or mobile images linked to a sample. Preprocessing will check image readability and apply model-specific resizing and normalisation. A YOLO detector will be compared with a Faster R-CNN baseline for the agreed localisable classes [5], [6].

The service will return labels, bounding boxes, confidence scores, image-quality flags and its model version. Diffuse appearance changes such as discolouration will use an agreed whole-image label or region annotation instead of an arbitrary object box. Severity will be reported only when a defensible scoring rubric has been validated. These outputs describe visible evidence for the integrated report.

3.4 Component 2 summary

Proposed owner: Component 2 student, identity pending. Inputs include pH, turbidity, hardness, total dissolved solids, fluoride, conductivity and dissolved oxygen where available. Each measurement will retain its unit, method and timestamp. The selected dataset must have a documented quality label and intended use.

Random Forest, an ensemble of tree predictors [7], and Gradient Boosting will be evaluated against a transparent parameter-rule baseline. Missing-value handling will be fitted using training data only. Tree-based models do not require routine scaling; any preprocessing will match the chosen estimator. Outputs will include the predicted category, an appropriately evaluated confidence measure, parameter flags and input-completeness status. Rule-based flags and learned classifications will remain separately visible.

3.5 Component 3 summary

Proposed owner: Component 3 student, identity pending. This component will analyse microscope images acquired using a recorded magnification and sample-preparation protocol. Experts will define the achievable categories, annotate organism locations and provide reference labels. Candidate methods include YOLO and Faster R-CNN with settings suited to small objects.

Outputs will include validated organism categories, locations, detection confidence and counts per image or field of view. Counts will not be converted to organisms per unit volume without a calibrated sampling method. The specific microorganism categories and any species-level claims must be agreed by expert reviewers and validated against reference methods; the supplied overview does not establish species-identification capability. Uncertain images will be referred for review.

3.6 Component 4 summary

Proposed owner: Component 4 student, identity pending. This component will combine the three analytical outputs into a plain-language report. Retrieval-augmented generation joins language generation with access to retrieved information

[8]. Here, retrieval will use a curated, versioned collection of water quality and treatment references, beginning with authoritative guidance [3].

Reports will identify findings, uncertainties, missing tests, supporting passages and proposed next actions. The language model will not set safety thresholds or invent treatment quantities. Any future dosage function will be implemented separately using reviewed formulas, explicit units and validated input conditions; unsupported cases will return a request for expert review. This report does not specify an operational chemical dosing procedure.

3.7 End to end user and data flow

A trained user creates a sample record and records collection time, source context and available observations. The user uploads the visible image and microscope images and enters or imports measurements. Validation checks file types, identifiers, required metadata and units before creating independent analysis jobs.

When jobs finish, the aggregation layer records their status and reconciles them under the same sample identifier and analysis version. It applies approved interpretation rules without averaging away conflicting evidence. Component 4 retrieves relevant passages and prepares the report. A reviewer can inspect the evidence, request new inputs and approve an updated report version. Later measurements create a new assessment snapshot rather than silently altering an existing report.

3.8 Data Acquisition, Sample Identity and Provenance

A sample should be assigned a globally unique, non-reused identifier before any analysis begins. The record should hold collection date and time, broad source category, collection and storage protocol, operator role, acquisition instrument, image reference, parameter units and available external reference results. Separate records are required for repeated collections because water properties change over time. Multiple images of the same sample must remain grouped and retain their relationship to one original water sample.

A common manifest is required for cross-modal testing. A row should describe the sample ID, each modality's presence, linked file IDs, the instrument or camera used,

acquisition session, quality checks and any known missingness. An image analysis request should not be able to attach findings to a numerically similar but different sample. This invariant should be enforced at upload, inference, aggregation and report generation. Checksums allow the research team to detect unintended replacement or duplication of image assets.

A training record should also identify the provenance and scope of every label. For contamination, the annotation source and visible region must be recorded. For numerical quality, the target class requires a documented reference laboratory measurement or explicitly justified rule definition. For microscopy, expert-validated organism annotations should carry the identification level, review decision and acquisition metadata. Dataset documentation should describe composition and intended limits in a datasheet-style record [10].

3.9 Label Taxonomy and Annotation Governance

The first annotation workshop should develop a label dictionary with inclusion criteria, exclusion criteria and examples. In visible imagery, “floating debris” and “surface oil film” may be candidates when experts can identify a reproducible image region; diffuse turbidity and general colour changes should be handled through a documented whole-image or segmentation protocol instead of forcing arbitrary boxes. Ambiguous instances should receive an uncertainty mark or an exclusion reason, not an invented definitive target.

Every bounding box must use a declared coordinate convention, such as x-min, y-min, x-max, y-max in original-image pixel coordinates. Both width and height must be positive and within the stored image dimensions. Inter-annotator agreement should be assessed on a reviewed subset; disagreements should be resolved by an adjudicator using recorded reasons. A separate quality-control sample should estimate whether labels remain consistent after changes to the taxonomy. Any altered policy must trigger versioning and reevaluation of affected records.

For microscope imagery, annotators must not assign a specific bacterial species solely because its appearance superficially resembles a database example. Species-level claims may require culture, biochemical or molecular reference evidence not available in ordinary images. Annotation should therefore operate at the finest level

justified by the independent standard and identify unresolved cases. The final thesis must state this evidentiary limit alongside detector performance rather than treating confidence scores as microbiological confirmation.

3.10 Dataset Partitioning and Leakage Prevention

The recommended evaluation partition is grouped by physical sample and collection event. Multiple photographs, augmentations or microscope fields from the same sample may resemble one another closely and must never be separated across training and held-out testing. Where data permit, an additional site- or session-held-out evaluation will assess transfer to new acquisition conditions. A split manifest with random seed and sample IDs should be fixed before hyperparameter tuning.

An independent final test set is used only once for the selected model configuration. Training-fold statistics should determine imputers and any scaling parameters, preventing information from test cases from leaking into preprocessing. Augmentation is applied to training data only. Class distributions and group sizes must be reported for each partition; rare classes warrant explicit uncertainty statements and potentially repeated cross-validation, without repeated tuning to the final holdout.

Combined-system evaluation requires physically or procedurally linked multi-modal examples. Merely attaching a water photograph to an unrelated tabular row and microscope image would create a demonstration scenario, not a valid sample-level study. Such artificial records may be useful for API contract verification if labelled synthetic, but they cannot support claims about integrated diagnostic performance.

3.11 Model Selection and Baseline Comparisons

For Components 1 and 3, a common object-detection evaluation protocol should compare a YOLO-family model and a Faster R-CNN-family baseline while holding label definitions and test cases constant [5], [6]. Report model version, training image counts, resolution, confidence threshold, non-maximum suppression settings, inference hardware and preprocessing. The purpose of comparison is to assess the accuracy–latency–resource trade-off; it does not establish that either family is universally superior.

For Component 2, compare Random Forest and Gradient Boosting with transparent parameter rules. A Random Forest produces an ensemble of decorrelated trees and can represent nonlinear interactions [7], but predictive success remains sensitive to label quality, training representation and class imbalance. Evaluation should include class-wise precision, recall, macro-F1 and calibration analysis where probabilistic outputs are used. Unsafe-class recall merits explicit emphasis because the consequences of a missed unsafe condition differ from a false warning.

For Component 4, compare three reporting approaches: a deterministic template populated with structured findings, an unrestricted language-generation baseline used only in a controlled evaluation environment, and a retrieval-supported and checked report. The scoring rubric should record claim correctness, citation support, missing-data honesty, inappropriate numerical treatment instructions, time per report and reviewer effort. A comparison of plausible fluency alone would not demonstrate reliable advice.

3.12 Integrated Decision Semantics

The aggregator receives typed component outputs; it must not reduce them to an average probability. Object detector confidence estimates are not calibrated probabilities of potability, and different models' confidence scales are not directly comparable. Instead, the shared record should hold a status for each stream, a set of findings, explicit parameter warnings, scope flags and provenance links. A rules engine applies review-approved policies to the structured results while preserving conflicts.

The statuses "concern detected," "no concern in assessed scope," "incomplete," and "review required" give reports consistent semantics. If a mandatory measurement is absent, the overall assessment must not silently default to the favourable state. If a visible detector reports an oil-like film while numerical results appear ordinary, both outputs should remain visible and the case should be escalated rather than overwritten. Re-analysis produces a new snapshot linked to the prior one and should not alter an already reviewed report.

Figure 3.2 provides the proposed input-to-report workflow, and Figure 3.3 shows the evidence entities needed to reproduce an assessment. These diagrams elaborate the

supplied four-component system design [1], [2]; they do not establish that the services have been implemented.

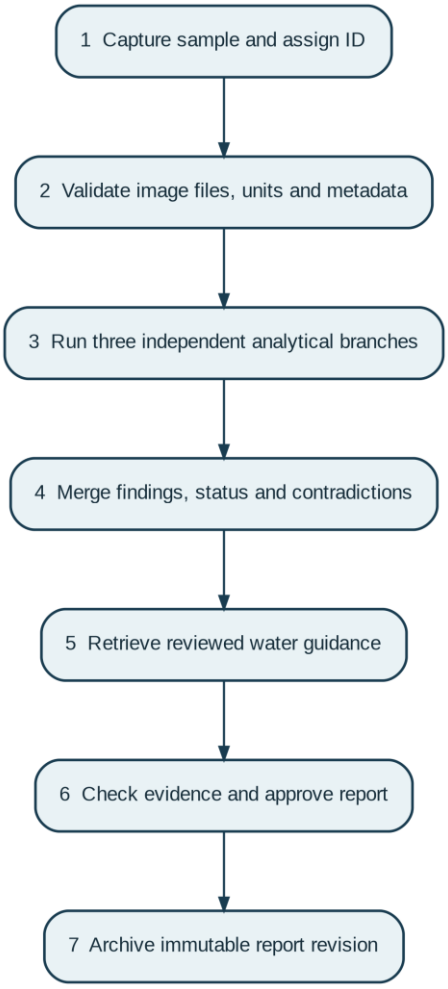


Figure 3.2: Integrated sample lifecycle and report release (proposed design; not experimental evidence).

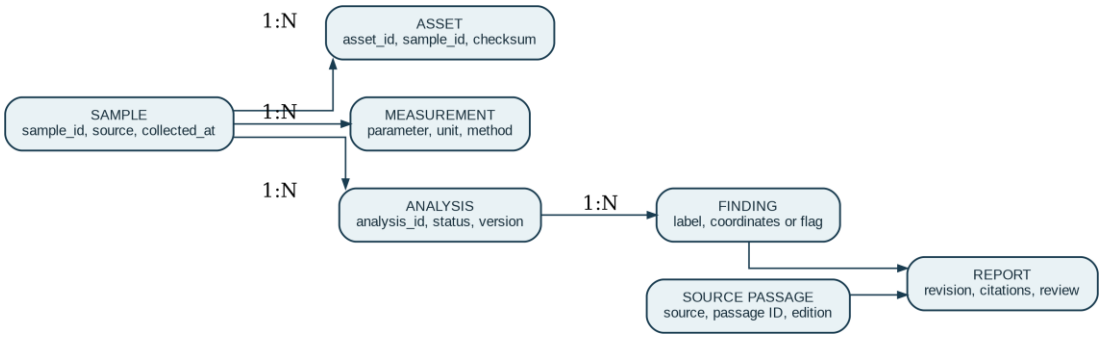


Figure 3.3: Logical data entities and evidence relationships (proposed design; not experimental evidence).

3.13 Retrieval, Reporting and Chemical Calculation Controls

The retrieval knowledge base should begin with recognised water quality and treatment guidance and only include documents for which source title, publisher, edition, URL or stable identifier, and review date can be retained [3], [9]. Text should be chunked with passage identifiers so every report claim can point to a specific reviewed segment. The reporting service receives only necessary structured facts and retrieved passages, not unrestricted web pages or instructions embedded in sample metadata.

The report generation prompt should separate system rules, structured analytical findings and untrusted retrieved content. Content that tells the model to bypass prior instructions is treated as data, not as authority. Generated references must be checked against retrieved passage identifiers, and unsupported statements should be removed or assigned a “requires professional review” status. When retrieval fails, the service should return a limited evidence summary rather than fabricating a citation or treatment procedure.

Any chemical calculator should use deterministic, unit-checked expressions maintained outside the language model. Inputs must include required quantities and treatment assumptions, and outputs should be blocked where those assumptions do not hold. The group report does not prescribe an operational chemical dose: the actual formulas, allowed ranges, reference sources, validation records and approval process must be documented and checked by an appropriate expert before operational use. Figure 3.4 outlines the proposed checks.

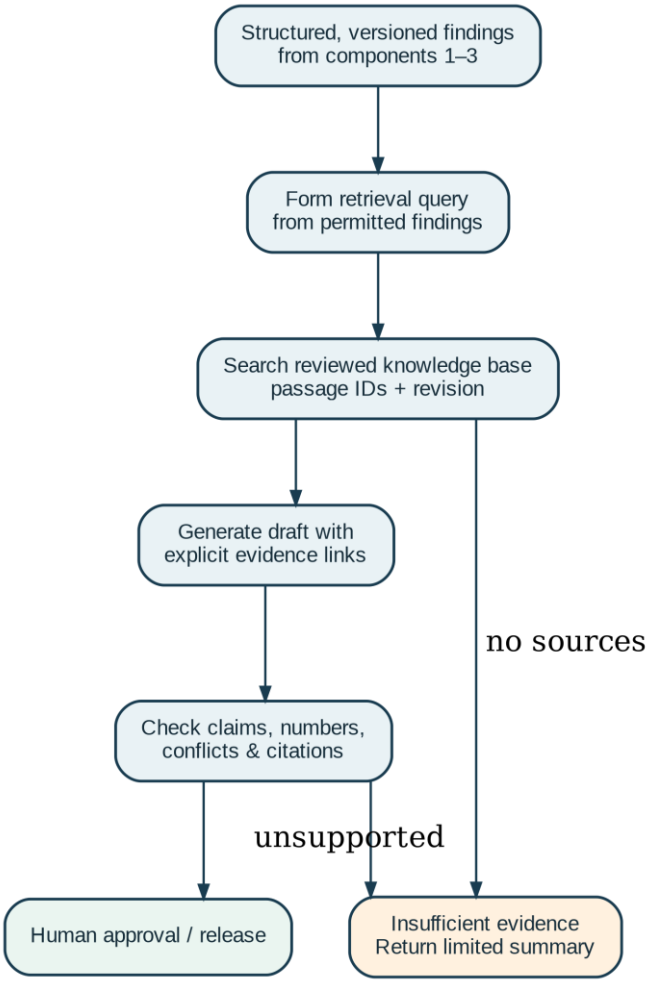


Figure 3.4: Retrieval-supported reporting with claim-checking safeguards (proposed design; not experimental evidence).

4 COMPONENT INTEGRATION AND COHESIVE OPERATION

4.1 Integration strategy

Integration will begin with a shared schema and mock responses so each component can be developed independently. The team will first connect upload and sample storage, then add Components 1 to 3, the aggregation logic and Component 4. Each service will own input validation and explicit status reporting; the integration owner will maintain sample association and the end-to-end workflow.

Development, test and pilot environments will use separate data and credentials. Interface changes will be versioned and reviewed jointly by the producing and consuming component owners. A small set of representative complete, incomplete and conflicting samples will form the integration regression set. The division of responsibilities is shown in Table 4.1.

4.2 Interfaces data contracts and shared models

The proposed REST interface will exchange JSON metadata and structured results; uploaded image bytes will be stored through a controlled file endpoint. Every result must contain sample_id, analysis_id, component_id, schema_version, model_version, timestamps and a status. Successful responses contain typed findings, while rejected or failed analyses contain error codes and a readable explanation.

Measurements must include parameter name, numeric value, unit and method. Detections must reference the original image and give coordinates in a declared convention. A confidence field describes model output and must not be labelled as the probability that water is safe. Appendix A defines the draft contracts. The aggregation service rejects mixed sample IDs and incompatible schema versions.

4.3 Shared nonfunctional requirements

Proposed shared requirements include authenticated access, role-based permissions, encrypted transport, access logging and configurable retention. Sample locations and operator details will be minimised and access controlled. Backups and restoration will cover both metadata and linked images. Model and reference versions will be retained to make reports reproducible.

The interface will use explicit text labels alongside colour, readable contrast and keyboard-accessible controls. A preliminary performance target is a 95th-percentile report response time of at most 60 seconds after all valid inputs have been uploaded, measured on a declared test configuration. This excludes sample collection, laboratory processing and network upload. The team must validate or revise this target using actual workload measurements.

4.4 Integrated workflow and exception handling

The normal workflow validates inputs, queues analysis, aggregates results and generates a report with citations. A missing modality produces a partial assessment with its scope stated. An unreadable image requests recapture; an unsupported unit is rejected; and an unavailable service records a failure with a retry option. A retry uses an idempotency key so that duplicate requests do not create duplicate analyses.

Conflicting outputs produce a review-required status and preserve all findings. If retrieval cannot supply an approved supporting source, Component 4 will show the structured findings and explain that guidance is unavailable. No favourable overall status is generated when a required input or review is missing. Report text is checked against the structured findings before release.

4.5 Team contribution and dependency matrix

Table 4.1 allocates proposed responsibilities to component roles because student names and actual contribution records have not been supplied. The matrix is a work allocation, not a declaration of completed contributions. All four students will jointly review interfaces, integration tests and the common report. Appendix B records the confirmation still required.

Table 4.1 Proposed contribution and dependency matrix

Role	Primary work	Shared dependency
Component 1 student	Image labels, detector and image tests	Image schema and sample storage
Component 2 student	Tabular labels, classifier and parameter rules	Measurement schema and aggregation
Component 3 student	Microscopy protocol, detector and expert labels	Image metadata and reference validation

Role	Primary work	Shared dependency
Component 4 student	Retrieval, report generation and grounding tests	Results from Components 1 to 3
All four students	Integration, evaluation and common report	Versioned contracts and evidence records

4.6 API Contract and Cross-component Validation

Every asynchronous analysis must be linked to the same sample_id and to a specific analysis revision. The envelope should include schema_version, component_id, model_version, request time, completion time, status and source asset identifiers. An implementation should validate required fields, type constraints, coordinate bounds and value units at the service boundary. Error responses must give stable machine-readable codes rather than silently returning an empty list that could be mistaken for a normal negative finding.

One example contract rule is that a measurement of "4.2" is not interpretable without naming its parameter and units; an unsigned value cannot be accepted as either pH or dissolved solids. A second is that an image bounding box must reference an actual stored image and fall within its pixel dimensions. Third, any report combining analysis IDs from different samples or incompatible schema versions should be rejected. These requirements are testable independently of model accuracy.

The team should maintain a contract repository with approved examples for successful detection, no detection, incomplete input, model rejection and service failure. Contract tests should be run on each component release and should include backward compatibility checks. Interface version changes must be discussed by the affected component owners before deployment. A machine-readable schema such as JSON Schema can help distinguish syntactic conformance from additional contextual domain validation.

Table 4.2: Proposed shared API contract fields

Field	Example type	Validation purpose
sample_id	UUID/string	Associates all four branches with one sample
analysis_id	UUID/string	Distinguishes repeated model runs
schema_version	Semantic version	Rejects incompatible payloads
component_id	Enum 1–4	Preserves analytical provenance

Field	Example type	Validation purpose
source_asset_id	String	Identifies analysed image or record
status	Enum	Differentiates failed, missing and negative findings
model_version	String	Supports reproducibility and rollbacks
created_at	ISO 8601 time	Records assessment order
findings	Typed collection	Preserves category, warnings and coordinates

4.7 Orchestration, Retry and Failure State Design

The orchestration service should treat each component run as a separately observable job. A request transitions from received to validated, processing and one of completed, rejected or failed; a complete sample may later enter review or release states. The service must avoid interpreting a network timeout as a negative analysis. A bounded retry policy, idempotency key and immutable run ID allow recoverable failures to be retried without duplicating sample records or overwriting previous evidence.

A partial sample is a legitimate input only when the application clearly declares the narrower scope of assessment. The report must name every unavailable modality and explain why findings are limited. If a model returns an out-of-taxonomy class, low-quality acquisition flag, contradictory parameter result or unsupported dosing request, the aggregator should route the record for review. A reviewed report should include an immutable snapshot of inputs, model versions and knowledge-base revision.

Figure 4.1 defines the proposed state transitions. It is intended as a verification aid: each arrow can be associated with a test case, expected next status and audit event. The implementation team should validate this model using fault injection for service unavailability, malformed values, duplicate requests and cross-sample ID substitutions.

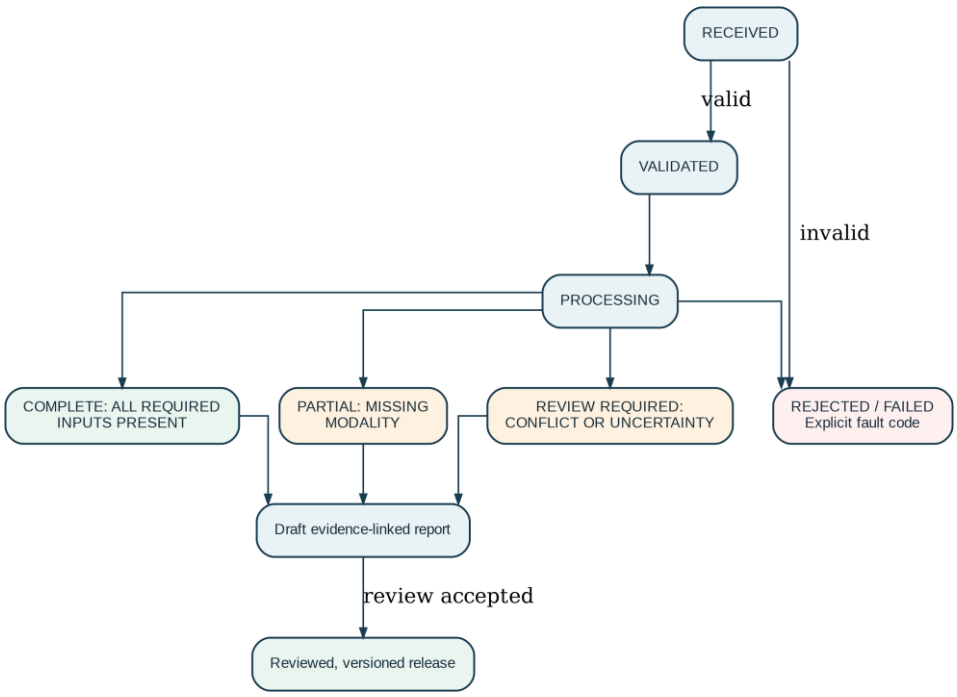


Figure 4.1: Analysis state machine and fail-safe routing (proposed design; not experimental evidence).

4.8 Auditability and Reproducibility Requirements

The system should maintain an append-only audit trail for creation, update, analysis start, status changes, report review and report release. Each event records actor role, sample ID, relevant object identifier, time and action without unnecessarily exposing personal or location details. The release manifest must make it possible to reproduce an assessment using saved inputs and exact preprocessing, model, rules and reference versions.

Versioning should apply independently to label taxonomies, datasets, trained weights, API schemas, rule configurations and the retrieval knowledge base. Model cards should record intended use, data representation, key evaluation metrics and limitations [11]. Research artefacts should include dataset split manifests, experiment configuration files, evaluation scripts, test logs and evidence records. Missing items should be explicitly reported rather than filled with inferred development histories.

5 INTEGRATED IMPLEMENTATION AND PROJECT EXECUTION

5.1 Common technology stack and infrastructure

The proposed stack uses Python for preprocessing and inference, PyTorch for image models, and scikit-learn-compatible tooling for tabular classifiers. A REST service will expose analysis and reporting functions, while a browser interface will support sample entry, evidence inspection and report viewing. The precise frameworks and versions will be selected and recorded during implementation.

PostgreSQL is proposed for sample metadata and structured results, with controlled object storage for images and model files. Retrieval will use a searchable index containing passage identifiers, source metadata and approved versions. A shared repository will contain contracts, orchestration, deployment configuration and integration tests; model experiments may be organised in component directories. These are proposed choices, not an inventory of an implemented deployment.

5.2 Development and integration process

Work will proceed through requirements confirmation, dataset preparation, component baselines, interface integration and integrated evaluation. Short iterations will end with a demonstrable increment and a recorded review. Pull requests will identify the changed interface, expected behaviour and validation evidence. Continuous integration will run schema checks, unit tests for deterministic rules and the integration regression set.

The team will freeze held-out test data before model selection. Training configurations, seeds, dataset versions and evaluation scripts will be retained with each model release. A model or knowledge-base change will trigger report regression tests because it may alter user-facing findings without changing the interface.

5.3 Deployment and operational configuration

The proposed deployment packages the interface, API, inference workers and reporting service as separately configurable services. Secrets will be supplied through environment-specific secret management. Each release will identify the application revision, model artefacts, schema version and approved knowledge-base snapshot.

Monitoring will record job failures, queue length, response time and unusual input distributions without logging unnecessary personal information. Operational procedures will define retry limits, service health checks, backup schedules and a restoration drill. A previous compatible model and application release will remain available for rollback. Deployment, recovery and monitoring have not yet been demonstrated in the supplied evidence.

5.4 Evidence of team execution

The supplied materials contain functional descriptions but no Git history, issue records, meeting minutes, MS Planner exports, Teams records or research diaries. Therefore, this report cannot verify development dates, task completion, workload distribution or team attendance.

For the final submission, the team should link each milestone to a repository revision, reviewed issue, experiment record and meeting decision where applicable. Evidence should show both component work and shared integration changes. Appendix B provides the proposed contribution categories to reconcile against these records before the students sign their declaration.

5.5 Development Framework and Verification Strategy

A staged, iterative process is appropriate because the research components have different data and infrastructure dependencies. The first iteration should establish the common sample schema and mock API responses. The second should demonstrate upload, measurement and image validation. The third should connect the first analytical components using fixed fixtures before attempting full model integration. Later iterations should incorporate retrieval, reporting, access control, deployment and supervised pilot evaluation.

Each increment should finish with a demonstration, repository tag, contract-test report and documented decisions. A small regression dataset should include complete samples, missing measurements, unreadable images, unexpected labels, simulated service failures and conflicting analyses. The integration test suite should run automatically when schemas, models, rules or retrieval content change. Without

these records, the report must describe intended verification rather than claiming continuous integration was conducted.

The final thesis should provide evidence linking each requirement to a design element and a test. Requirements coverage is especially important because an attractive interface can appear complete while omission of error handling undermines safety. The ISO/IEC 25010 product quality model offers a useful vocabulary for evaluating reliability, usability, security, maintainability and related product qualities [12].

5.6 Deployment Topology and Operational Security

The reference deployment separates the interface, API layer, image inference workers, tabular inference service, aggregation service and RAG generation service. Image assets and metadata are stored with different controls, and the reference library is versioned independently. In a low-resource pilot, several software services may share one host while retaining their logical boundaries. The selected hosting environment, GPU specification, inference runtime and data retention configuration must be recorded as part of the actual implementation.

Security requirements include transport encryption, role-based access, validated file types, upload size limits, server-side authorisation for every sample, controlled credentials, separation of development and production, and backups. File extensions alone are not sufficient to assess uploaded content. Retrieved passages and user-provided sample descriptions should be considered untrusted inputs that cannot redefine the reporting system’s rules. Access attempts, prompt-injection tests and unexpected data exposure must be included in security validation.

Operational observability should cover job queue time, inference duration, failure count, retries, model load errors, retrieval failures and report approval latency. Reliability testing must specify workload and hardware rather than treating a single favourable run as general performance. A practical rollback plan retains the previous validated release while keeping a record of which version produced each user-visible report.

5.7 Professional Practice, Accessibility and Ethics

Potentially sensitive data include exact sampling locations, photographs containing people or identifying objects, operator details and water-source ownership information. The default data model should collect only what is necessary, restrict access by role and document retention. Participant involvement in usability trials requires clear information about the study and consent procedures appropriate to the university's review requirements. Water samples, laboratory images and environmental records also need provenance and permission to reuse in publications.

The interface should not rely only on red and green indicators; textual statuses, warning explanations and keyboard navigation support wider accessibility. Reports must show assessment scope, acquisition timestamp, confidence limitations and the difference between screening results and professional certification. The NIST AI Risk Management Framework provides an approach for identifying, measuring and managing AI risk throughout development [13]. In this project, such risk management should address domain shift, false reassurance, unsupported advice, labelling uncertainty, security and model drift.

The proposed study should avoid producing treatment advice that might reasonably be followed without adequate expertise. A human-review mechanism and explicit warning about unmeasured hazards help limit that risk, but these design controls require empirical validation. Any live pilot should be supervised by qualified water-quality personnel and should not replace accredited laboratory tests or public-health decisions.

6 INTEGRATED TESTING EVALUATION AND VALIDATION

6.1 Evaluation framework and success criteria

Evaluation will ask whether the combined system preserves evidence accurately, detects the agreed classes, reduces unsupported report claims and helps intended users interpret results. Component tests will use unseen examples, and integrated tests will use sample records containing linked modalities with an independent reference assessment. A collection of unrelated public image and tabular datasets may support component development, but it cannot by itself validate sample-level fusion.

Data will be partitioned by original sample and, where possible, site or acquisition session to prevent near-duplicate leakage. The test set will be held out from model and threshold selection. The team will report class counts, confidence intervals and results by acquisition condition. Reference labels and uncertain cases will be reviewed by qualified annotators.

Provisional engineering acceptance criteria are complete sample association and provenance in all scripted tests; rejection of every deliberately malformed contract case; zero unsupported numerical treatment instructions in the curated report test set; and the latency target in Section 4.3. Model acceptance thresholds must be agreed with supervisors and domain reviewers before final testing. These are proposed criteria, not achieved results.

6.2 Integration and end to end testing

Testing will cover schema validation, service integration, complete and partial samples, duplicated requests, timeouts, conflicting results, unauthorised access and report revisions. Security tests will include access to another user's sample and attempts to place instructions inside uploaded or retrieved content. Regression tests will verify that previously resolved faults do not return.

The evaluation will compare individual reports, a rules-only combined summary and the full retrieval-supported workflow. Measures will include missed warnings, unsupported claims, report completion rate, review time and user comprehension.

Appendix C lists concrete integrated test cases and their expected behaviour. Execution dates, observed outcomes and defect records remain pending.

6.3 Overall results

No experimental or operational results were supplied. The current output is an integrated research design, including component boundaries, proposed data contracts, a reporting workflow and an evaluation plan. Accuracy, recall, processing speed, user satisfaction and treatment effectiveness cannot be reported as achieved.

Table 6.1 Required results and current evidence status

Evaluation area	Measures to report	Current status
Visible detection	Per-class recall and mAP	Not measured
Quality classification	Unsafe recall, macro F1 and calibration	Not measured
Microorganism detection	Per-class recall and count error	Not measured
Report quality	Citation support and expert-rated consistency	Not evaluated
Integration	Association errors, failures and latency	Not tested
User validation	Comprehension and task completion	Not conducted

Table 6.1 identifies the results that must be obtained before the report can support conclusions about performance. A completed report should add dataset sizes, test conditions, uncertainty estimates, representative failure cases and links to component experiments. Numerical examples must be clearly distinguished from measurements.

6.4 Stakeholder and user validation

Stakeholder validation is planned with intended operators and suitably qualified water quality reviewers. Participants will review complete, incomplete and contradictory sample cases and answer comprehension questions about the report's status, limitations and recommended next step. The team will record task completion, interpretation errors and qualitative concerns, then map findings to design changes.

Recruitment, participant count, consent arrangements and study approval requirements must be settled before data collection. No interviews, user acceptance

sessions or expert approvals are claimed in this draft. Reports should be tested in the languages and literacy contexts selected for the actual deployment setting.

6.5 Achievement of common objectives

O1, O2 and O3 have been specified as component research objectives, but their model performance has not been demonstrated. O4 has a reporting and retrieval design, but grounding and expert acceptability remain untested. O5 has a proposed integration architecture and test matrix, but no executed end-to-end evidence.

The shared design work supports planning across all five objectives. None should be marked fully achieved on the basis of the supplied descriptions alone. Final achievement decisions must reference the held-out results, integration logs and stakeholder evidence identified in Sections 6.1 to 6.4.

6.6 Integrated limitations and threats

Important validity threats include small or imbalanced datasets, repeated images from the same sample, inconsistent labels and changes in cameras, microscopes or sampling conditions. A model may perform well on one dataset yet fail when the collection protocol changes. Component confidence scores are not automatically comparable, so averaging them would not establish a valid overall safety score.

The integrated assessment is also limited by missing modalities and hazards outside the selected inputs. Retrieval can return irrelevant or outdated material, and generated explanations can misrepresent valid evidence. The design therefore requires versioned references, consistency checks and expert review. The proposed safeguards themselves need evaluation; they do not prove that the system is reliable.

6.7 Component-level Evaluation Protocols

Component 1 evaluation should use a frozen set of photographs from held-out samples, with class-wise detection counts and ground-truth regions. Report precision, recall, F1 and mean average precision for declared intersection-over-union thresholds, together with image-level analysis of diffuse labels when applicable. Aggregate scores must be supported by example images of false positives and missed objects, acquisition-condition breakdowns and the actual number of test samples.

Training screenshots or validation-set metrics cannot substitute for a final held-out assessment.

Component 2 evaluation should use a documented quality label and units for every input parameter. Report the class distribution, confusion matrix, macro-F1, balanced accuracy and sensitivity for any defined unsafe category. Compare the selected ML estimator against a fixed rules-only baseline using identical test records. Calibration should be assessed before interpreting a probability output as a decision-support confidence value; predictions near a threshold must be handled according to a pre-specified review rule.

Component 3 evaluation should distinguish object-localisation quality from species identification. Expert-verified targets should define the supported label level, and counts should be evaluated only when the image field of view permits meaningful counting. Results should be stratified by magnification, image quality, staining and acquisition site when sufficient data are available. Inter-rater disagreement and uncertain targets must appear in the error analysis. The research cannot claim laboratory confirmation of organisms solely from visual prediction.

Component 4 evaluation should score factual consistency with component outputs, citation validity, coverage of mandatory warnings, the ability to acknowledge missing inputs, readability and compliance with the deterministic dosing guardrails. Reviewers should mark each factual claim as supported, unsupported, contradicted or outside scope. Measure unsupported-claim rate as unsupported claims divided by assessed factual claims, and citation precision as supported citations divided by evaluated citations. A report that is fluent but unsupported must not be scored as successful.

6.8 Integrated Validation and Baseline Comparison

The full-system evaluation must test linked samples, not merely aggregate independent component benchmark scores. Each integrated test item should contain a sample ID, available modalities, reference interpretation, acquisition metadata and expected handling of missing or contradictory evidence. The evaluation should compare the proposed pipeline against a basic display of separate component outputs

and a deterministic template. Its purpose is to determine whether evidence aggregation and retrieval-supported explanation provide additional practical value.

Outcome measures include association error rate, missing-warning rate, incorrect favourable-status rate, reviewer time per case, report factual support, status accuracy and total processing latency. Test fixtures should deliberately include swapped sample IDs, duplicate uploads, outdated schema versions, impossible units and absent reference passages. Since the user-facing conclusion depends on all four branches, a high individual model score does not guarantee low system-level error.

For a controlled stakeholder assessment, independent reviewers should be presented with matched cases in counterbalanced report formats. The evaluation form should capture accuracy of interpretation, time to identify primary concerns, perceived clarity, confidence in traceability and willingness to escalate incomplete results. Such measurements require ethical review and actual study records; no completed stakeholder evaluation is claimed by this design draft.



Figure 6.1: Proposed grouped evaluation and validation protocol (proposed design; not experimental evidence).

6.9 Statistical Reporting and Robustness Analysis

Where appropriate, uncertainty should be reported through confidence intervals calculated over independent samples or sample groups. Comparing models on the same held-out cases permits paired analysis of errors, but repeated crops from one original sample must not be treated as independent observations. Metrics should be interpreted alongside the underlying class counts, label disagreement and data-collection limitations. Models with very small test classes need explicit caution in claiming generalisation.

The robustness plan includes variations in camera lighting, reflection, image resolution, turbidity levels, sensor drift, missing numerical fields, microscope magnification and background artefacts. Performance should be documented across the variations present in the actual dataset. If conditions are not covered, the final

report should state that robustness remains untested instead of extrapolating from ordinary samples.

Threshold analysis should be kept separate from final test evaluation. Any adjustment to confidence thresholds, unsafe decision rules or review conditions must be selected using development data before evaluation on the locked test set. Post-hoc tuning against the test set invalidates the claim of independent generalisation and must be disclosed if it occurs.

6.10 Results Reporting and Evidence Register

The current uploaded common report establishes proposed architecture, component responsibilities, interfaces and test designs. It does not provide measured confusion matrices, trained model artefacts, sample manifests, test-case logs or expert review scores. Therefore no value should be entered in the final thesis as an achieved accuracy, latency, detection count or treatment efficacy unless its original supporting record can be produced. Table 6.2 lists the minimum evidence needed for the final results chapter.

The completed research report should distinguish results that address the scientific question from software verification findings. A high unit-test pass rate can show that identifiers and bounding-box conventions are preserved, but it does not demonstrate that images are classified correctly. Similarly, a high object-detection score does not demonstrate that a report faithfully states which measurements are missing. Different measures must be reported with their corresponding unit of analysis and relevant reference standard.

After results are supplied, Section 6.3 should be updated with actual numerical outcomes and figures, Section 6.5 with objective-by-objective achievement, and the discussion with both favourable and adverse findings. Any discrepancy between the proposal and completed implementation should be documented rather than silently replacing the historical project plan.

Table 6.2: Required empirical evidence before final submission

Component/area	Required artefact	Status in supplied files
Visible detection	Test split, class AP/recall, model weights, error examples	Not supplied

Component/area	Required artefact	Status in supplied files
Tabular classification	Data dictionary, label rule, confusion matrix, unsafe recall	Not supplied
Microscopy	Expert labels, acquisition details, detections, counts	Not supplied
RAG advisor	Approved corpus, claim-level review and citations	Not supplied
Chemical calculator	Formula references, input bounds, unit tests, approval	Not supplied
End-to-end	Linked sample manifests and executed integration tests	Not supplied
Stakeholders	Consent, reviewer rubric, anonymised responses	Not supplied
Project execution	Repository history, roles, release logs	Not supplied

7 COMMERCIALISATION SUSTAINABILITY AND RESPONSIBLE INNOVATION

7.1 Value proposition and beneficiaries

The proposed value is a single report that links visible, measured and microscopic evidence to its source. Laboratory teams could spend less time assembling reports, while monitoring officers could review incomplete or conflicting samples more consistently. Potential beneficiaries include small water service operators and research laboratories with access to the required instruments.

These are value hypotheses. The team has not supplied stakeholder evidence showing demand, willingness to pay, time savings or better decisions. A pilot should measure preparation time and interpretation errors against the user's existing reporting process before making commercial claims.

7.2 Market alternatives and differentiation

Relevant alternatives include laboratory reports, parameter-threshold spreadsheets, sensor dashboards, stand-alone image classifiers and manual expert interpretation. Each addresses part of the workflow, and a laboratory service may remain the appropriate route for confirmatory testing.

The proposed differentiation is the association of multiple modalities with one sample and the traceability of report statements to evidence and references. A current competitor review and user comparison are required to establish whether this provides a practical advantage. No exclusive market position or unique technical capability is claimed.

7.3 Business model and feasibility

An initial service model could provide an institution with sample management, analysis and report generation through a subscription or supported on-premises installation. Research or public-sector partnerships may offer an alternative sustainability route. Early partners would ideally provide laboratory expertise, representative samples and feedback on operational fit.

Costs include data collection, annotation, instrument access, model development, hosting, generation usage, security maintenance and expert review. Feasibility should be measured using total cost per reviewed sample, including the cost of required measurements. Pricing, revenue projections and break-even figures remain unestimated because usage volumes and operating costs have not been supplied.

7.4 Adoption intellectual property and compliance

Adoption depends on reliable input collection, affordable instrument access, clear explanations and confidence in the review process. Training materials should show how to collect valid inputs and interpret incomplete findings. The team should document ownership of code and annotations and check licences for datasets, pretrained models and reference documents before redistribution or commercial use.

Before a pilot, the institution should review applicable data protection obligations, research ethics requirements and responsibilities for water safety decisions in the deployment setting. No regulatory approval or legal compliance is asserted here. The prototype will retain a human review step and will not automatically control chemical treatment.

7.5 Sustainable Development Goal contribution

The intended connection to SDG 6 is improved access to interpretable monitoring information, consistent with the focus of targets 6.1 and 6.3 [4]. Evaluation should measure practical indicators such as usable reports, interpretation errors, time to follow-up and the ability to monitor underserved sites. These project measures must not be presented as official national SDG indicators.

Possible negative effects include false reassurance, unequal access to instruments, unnecessary repeat testing and computing costs. Sustained benefit would require affordable workflows, local technical support, periodic validation and a clear route from an identified concern to competent follow-up. Population-level impact remains a future research question.

7.6 Commercialisation Validation and Positioning

The initial commercialisable proposition is an evidence-linked decision-support application for trained environmental monitoring users, not a consumer device that

certifies water as drinkable. Its differentiation must be supported by observation of user workflow and comparison against actual competing testing, logging and laboratory information systems. The research should document who pays, who acts on a recommendation, what laboratory confirmation remains necessary and what forms of liability or regulation could affect deployment.

A discovery pilot should test whether the system reduces time spent collating three forms of evidence and whether users can identify missing or contradictory information more reliably. Interviews with laboratory staff, small water utilities or environmental teams could explore data-import needs, instrument compatibility and acceptable review workflows. Survey conclusions must be based on genuine recorded responses rather than marketing assumptions. For a deployment decision, operating costs and validation costs matter as much as the expense of the inference models themselves.

Commercialisation milestones should include technical feasibility, expert validation, pilot usability, security review, data-governance review and stakeholder willingness to pay. Deployment and scale can be considered only after these stages are documented. Claims about health outcomes, cost savings or market size are hypotheses at the proposal stage.

7.7 Preliminary Sustainability and Cost Framework

A suitable total-cost framework accounts for image annotation, domain expert time, laboratory labels, cloud inference, data storage, retrieval indexing, user support, software maintenance and periodic model revalidation. GPU inference may dominate computational expense in heavy imaging workloads, but human review and reliable reference data may dominate the actual research budget. A final cost-benefit analysis should use measured usage and local procurement quotations, recording their dates and assumptions.

Sustainability has operational and social dimensions. A system that produces warnings users cannot act on is unlikely to improve outcomes. Likewise, a model that requires frequent retraining without a validated dataset process creates maintenance risk. The proposed modular architecture allows components to be upgraded independently, but every replacement must trigger cross-component

contract and regression tests. Contribution to SDG 6 remains a plausible pathway, not an observed reduction in contamination or increase in safely managed drinking-water access [4].

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Overall conclusion

The proposed solution addresses the common problem by joining three distinct evidence streams and an evidence-linked reporting service around one water sample. The design explains how the components should interact, how incomplete results should be represented and what must be tested before the system can support operational decisions.

The available evidence supports the coherence of the proposed design, but it does not establish detection accuracy, reliable safety classification or effective purification advice. The research question remains open until representative sample-level evaluation and expert review have been completed.

8.2 Principal integrated contribution

The principal integrated contribution at this stage is a specification for connecting visible detections, numerical parameter assessments and microscopic findings to a traceable report. Shared identifiers, typed interfaces, preserved uncertainty and reference-linked explanations define how the four individual research components form one system.

Whether this integration improves user decisions or reporting efficiency is the main claim to investigate next. Demonstrating that improvement requires comparison with simpler reporting arrangements and transparent reporting of failures.

8.3 Future directions

First, confirm users, acquisition protocols and supported classes, and secure representative linked samples and reference labels.

Next, establish model baselines, implement interfaces and complete integrated testing and expert review. Field deployment, additional languages and sensor integration should follow validated performance.

8.4 Synthesis of Group-level Contributions

The research design contributes a four-branch architecture under one sample identity; a traceability contract for images, measurements, findings and reference passages;

fail-safe handling for missing or contradictory modalities; a guarded separation between language generation and quantitative treatment calculations; and an evaluation plan that tests scientific and software quality separately. These contributions are design artefacts until the implementation and measurements are supplied.

The group must distinguish each student's independently developed scientific work from the shared integration layer. The final common report should include a signed allocation confirmed by repository history, experiment records, supervisor review and evidence of cross-component collaboration. The absence of such records in the present materials is a reporting limitation and must be resolved before submitting declarations of work performed.

8.5 Prioritised Completion Plan

The first priority is to obtain verified technical documentation from each of the four component owners, including supported classes, datasets, measured outcomes, screenshots, model versions and source code or run records. The second is to freeze a linked validation cohort and execute contract, integration and reporting tests. The third is to conduct domain-expert review of uncertain detections, water-quality status language and retrieval-supported guidance. The final stage is to populate the thesis with measured results, reconcile student contributions and obtain authorised signatures.

These activities should be scheduled against the actual submission date and available laboratory capacity, neither of which is established in the present files. Once completed, the group can replace the explicit “not measured” statements with evidence-linked findings and revise the conclusion to discuss observed contributions rather than intended functionality. No unverified implementation success is assumed in this version.

REFERENCES

[1] Project team, “Automated Water Quality & Contamination Detection Using Computer Vision, Machine Learning, Deep Learning and GenAI — Project Component Overview,” supplied two-page project proposal, 2026.

[2] Project team, “Automated Water Quality & Contamination Detection: Proposed System Architecture,” supplied system architecture diagram, Oct. 2026.

[3] World Health Organization, Guidelines for Drinking-water Quality, 4th ed., incorporating the first, second and third addenda. Geneva: WHO, 2026. https://www.who.int/publications/i/item/9789240121225 (accessed Oct. 7, 2026).

[4] United Nations Department of Economic and Social Affairs, “Goal 6: Ensure availability and sustainable management of water and sanitation for all.” https://sdgs.un.org/goals/goal6 (accessed Oct. 7, 2026).

[5] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, “You Only Look Once: Unified, Real-Time Object Detection,” arXiv:1506.02640, 2015. https://arxiv.org/abs/1506.02640.

[6] S. Ren, K. He, R. Girshick, and J. Sun, “Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks,” arXiv:1506.01497, 2015. https://arxiv.org/abs/1506.01497.

[7] L. Breiman, “Random Forests,” Machine Learning, vol. 45, pp. 5–32, 2001, doi: 10.1023/A:1010933404324.

[8] P. Lewis et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” arXiv:2005.11401, 2020. https://arxiv.org/abs/2005.11401.

[9] World Health Organization, Evaluating Household Water Treatment Options: Health-Based Targets and Microbiological Performance Specifications. Geneva, Switzerland: WHO, 2011. [Online]. Available: https://www.who.int/publications/i/item/9789241548229

[10] T. Gebru et al., “Datasheets for Datasets,” Commun. ACM, vol. 64, no. 12, pp. 86–92, Dec. 2021. [Online]. Available: https://www.microsoft.com/en-us/research/publication/datasheets-for-datasets/

[11] M. Mitchell et al., “Model Cards for Model Reporting,” in Proc. Conf. Fairness, Accountability, and Transparency, 2019, pp. 220–229, doi: 10.1145/3287560.3287596.

[12] ISO/IEC, “Systems and software engineering — Systems and software Quality Requirements and Evaluation (SQuaRE) — Product quality model,” ISO/IEC 25010:2023, 2023. [Online]. Available: https://www.iso.org/standard/78176.html

[13] E. Tabassi, Artificial Intelligence Risk Management Framework (AI RMF 1.0), NIST AI 100-1. Gaithersburg, MD, USA: NIST, 2023, doi: 10.6028/NIST.AI.100-1.

APPENDICES

The following appendices define the draft interface, proposed contribution allocation and integrated validation cases referenced in Chapters 4 to 6. They are design artefacts; they do not represent completed tests or signed approvals.

Appendix A Integrated interface specification

The API contract will use schema version 1.0 initially. POST /samples creates a record; POST /samples/{sample_id}/inputs attaches validated inputs; POST /samples/{sample_id}/analyses starts processing; GET /analyses/{analysis_id} returns status; and GET /samples/{sample_id}/reports/{report_id} retrieves an immutable report version. These routes are proposed and have not been implemented or tested.

Required common fields are sample_id, analysis_id, component_id, schema_version, model_version, created_at and status. Status is one of pending, running, completed, rejected or failed. A completed analysis may return no detections, but a missing or failed analysis must not do so as a substitute for status.

Image detections contain image_id, class_id, confidence and bbox_xyxy in original-image pixel coordinates with a top-left origin. Bounds must lie inside the image dimensions, and the stored image must match the referenced checksum. Measurements contain parameter, value, unit, method, measured_at and quality_flags. Missing values use null with an explicit reason, never an invented zero.

Aggregation records assessment_status as concern_detected, no_concern_in_assessed_scope, incomplete or review_required. Conflict and missing-data flags are retained separately. The no-concern state requires all configured mandatory inputs, no active rule violation and no unresolved review flag; it is not a potability certificate. A report contains finding_ids, source_passage_ids, model_versions, knowledge_base_version and reviewer_status.

Validation errors return HTTP 422 with field-level details; an unknown sample returns 404; authentication or permission failures return 401 or 403; a conflicting revision returns 409; and a temporary service failure returns 503. Raw credentials

and sensitive internal details must not appear in errors. Server-side access checks apply to every route.

Appendix B Team contribution declaration

The proposed allocation assigns visible contamination detection to the Component 1 student, tabular water quality classification to the Component 2 student, microorganism detection to the Component 3 student, and reporting with retrieval to the Component 4 student. Shared work covers requirements, contracts, aggregation review, integrated testing, responsible-use review and the common report.

Names, student IDs, work periods and evidence links must be reconciled with actual contributions before signing. Each student should confirm their own contribution and the shared work. No supervisor certification, student signature or agreement is inferred from the proposed allocation.

Appendix C Integrated test matrix

All cases below are planned. Actual outcomes, execution dates and evidence links must be added after testing. Table C.1 defines expected behaviour rather than measured results.

Table C.1 Planned integrated test cases

Case	Input or event	Expected behaviour
T01	Complete valid sample	All outputs linked to the correct sample
T02	Microscope input missing	Partial scope shown; no full safety claim
T03	Image fails quality checks	Recapture requested; analysis marked rejected
T04	Unit or value invalid	Field rejected with a clear explanation
T05	Conflicting component results	All evidence preserved; review required
T06	Model service times out	Failure shown; retry does not duplicate work
T07	No approved retrieval passage	Findings shown; unsupported advice withheld
T08	Another user requests a sample	Access denied and event logged
T09	Repeated analysis request	Idempotency prevents duplicate analysis

Case	Input or event	Expected behaviour
T10	Instructions embedded in source	Source cannot override reporting controls
T11	Inputs change after report	New report version; previous record retained
T12	Representative workload	95th-percentile latency measured and reported

Appendix D: Shared Data Dictionary

The following schema is an implementation proposal, not a copy of a running database. It records minimum fields required to link analytical evidence and reconstruct the report. Component owners must approve field types and allowable values before freezing the first interface release.

Entity	Essential fields	Reason
Sample	sample_id, source_category, collected_at	Collection provenance
Asset	asset_id, sample_id, checksum, path	Image/sample integrity
Measurement	sample_id, parameter, value, unit, method	Typed physicochemical evidence
Analysis	analysis_id, component_id, model_version, status	Execution provenance
Detection	image_id, class_id, confidence, bbox_xyxy	Localised evidence
Knowledge passage	passage_id, publisher, edition, source_uri	Advice grounding
Report	report_id, analysis_snapshot, reviewer_status	Reproducible released output
Audit event	actor_role, event_time, event_type, object_id	Access and change accountability

Appendix E: Expanded Fault-Injection and Security Test Plan

Each test is planned; a real result requires an execution date, software revision, actual observation and tester. Injected faults should be run against an isolated test environment using synthetic non-sensitive input records.

Case ID	Injected condition	Expected response
SEC-01	Access sample belonging to another user	403, no data leakage
SEC-02	Instruction hidden in retrieved passage	Ignore instruction; evidence only
VAL-01	TDS value without unit	Reject with field-level explanation
VAL-02	Image box outside original frame	Reject invalid geometry
VAL-03	Mismatched sample IDs	Reject aggregation
RES-01	Microbiology job times out	Record failed/partial, never negative

Case ID	Injected condition	Expected response
RES-02	No retrieved guideline passage	Limited report, no fabricated citations
RES-03	Duplicate request with identical key	Return original job state
RES-04	Conflicting data/vision warnings	Retain both, require review
RES-05	Attempt unsupported dose inputs	Block computation and escalate
RES-06	New report after input correction	Create new immutable revision
RES-07	Model version changed	Trigger regression test suite

Appendix F: Dataset and Model Reporting Checklist

This checklist follows transparent research-documentation principles [10], [11]. It should be completed separately for each trained model and reviewed before a result is cited in the final thesis.

Artefact	Required details	Verification status
Data source and permission	Record actual project-specific evidence	Pending
Acquisition and image protocol	Record actual project-specific evidence	Pending
Label taxonomy and adjudication	Record actual project-specific evidence	Pending
Sampling units and class counts	Record actual project-specific evidence	Pending
Train/validation/test grouping	Record actual project-specific evidence	Pending
Model family and version	Record actual project-specific evidence	Pending
Hyperparameters and seeds	Record actual project-specific evidence	Pending
Software/hardware environment	Record actual project-specific evidence	Pending
Evaluation scripts and thresholds	Record actual project-specific evidence	Pending
Classwise results and confidence intervals	Record actual project-specific evidence	Pending
Failure and error analysis	Record actual project-specific evidence	Pending
Known scope limitations	Record actual project-specific evidence	Pending
Responsible-review contact	Record actual project-specific evidence	Pending

Appendix G: Academic Integrity and AI-use Disclosure

The supplied common report declares that generative AI assisted drafting and organising the project material. The four students must verify original research claims, source references, tests, figures and final prose. This draft has additionally used AI assistance to expand the literature framing and design explanations and to generate conceptual diagrams. It does not claim execution of the proposed

experiments. All use must be disclosed in the form required by the SLIIT module. Student signatures and supervisor certification remain pending.